Case Study 9: Fuel Consumption and Route Comparison

1. Problem Statement

Develop a Python program to calculate the fuel required and fuel cost for different routes. Compare the costs and identify the route with the lowest fuel cost.

2. Objectives

- To calculate fuel consumption for different routes.
- To calculate the total fuel cost.
- To compare route costs.
- To identify the cheapest route.

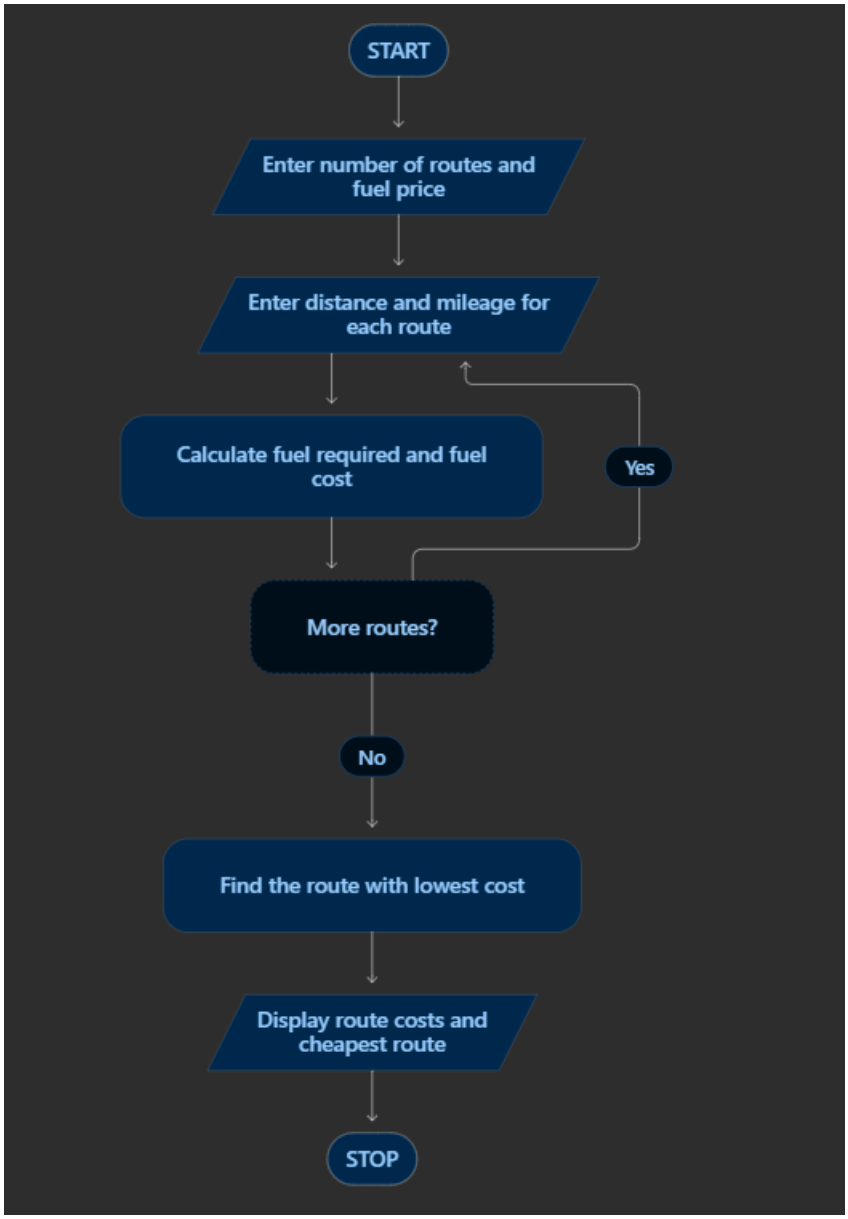
3. Assumptions

- The user enters the number of routes.
- The distance and mileage are entered for each route.
- Fuel price is entered by the user.
- Mileage is measured in kilometres per litre (km/L).
- The route with the lowest fuel cost is considered the cheapest.

4. Algorithm

1. Start.
2. Enter the number of routes and fuel price per litre.
3. For each route, enter its distance and vehicle mileage.
4. Calculate fuel required using distance divided by mileage.
5. Calculate fuel cost by multiplying fuel required by fuel price.
6. Display the fuel required and cost for each route.
7. Identify and display the route with the lowest cost.
8. Stop.

5. Flowchart



6.Pseudocode

START

INPUT number of routes

INPUT fuel price per litre

SET lowest cost to a large value

FOR each route

INPUT distance

```
    INPUT mileage
    CALCULATE fuel required = distance / mileage
    CALCULATE fuel cost = fuel required * fuel price
    DISPLAY fuel required and fuel cost

    IF fuel cost < lowest cost
      UPDATE lowest cost
      STORE route number
    END IF
END FOR

DISPLAY route with lowest cost
STOP
```

7.Source code

Fuel Consumption and Route Comparison.py

```python
n = int(input("Enter number of routes: "))
price = float(input("Enter fuel price per litre: "))

lowest_cost = float("inf")
cheapest_route = 0

for i in range(1, n + 1):
    print("\nRoute", i)

    distance = float(input("Enter distance in km: "))
    mileage = float(input("Enter mileage in km per litre: "))

    if distance < 0 or mileage <= 0:
        print("Invalid input!")
        continue

    fuel = distance / mileage
    cost = fuel * price

    print("Fuel required:", round(fuel, 2), "litres")
    print("Fuel cost: Rs.", round(cost, 2))

    if cost < lowest_cost:
        lowest_cost = cost
        cheapest_route = i

if cheapest_route != 0:
    print("\nCheapest route: Route", cheapest_route)
    print("Lowest fuel cost: Rs.", round(lowest_cost, 2))
else:
    print("\nNo valid route entered.")
```

8.Test cases

Test	Distance(km)	Mileage (km/L)	Fuel price	Expected fuel cost
Route 1	100	20	₹100/L	₹500
Route 2	120	15	₹100/L	₹800
Route 3	150	25	₹100/L	₹600

9.Output

```
"C:\Users\tejas\PycharmProjects\Python Activi
Enter number of routes: 1
Enter fuel price per litre: 100

Route 1
Enter distance in km: 100
Enter mileage in km per litre: 20
Fuel required: 5.0 litres
Fuel cost: Rs. 500.0

Cheapest route: Route 1
Lowest fuel cost: Rs. 500.0

Process finished with exit code 0
```

```
"C:\Users\tejas\PycharmProjects\Python Activit
Enter number of routes: 2
Enter fuel price per litre: 100

Route 1
Enter distance in km: 120
Enter mileage in km per litre: 15
Fuel required: 8.0 litres
Fuel cost: Rs. 800.0

Route 2
Enter distance in km: 110
Enter mileage in km per litre: 20
Fuel required: 5.5 litres
Fuel cost: Rs. 550.0

Cheapest route: Route 2
Lowest fuel cost: Rs. 550.0

Process finished with exit code 0
```

10.Conclusion

The program calculates fuel consumption and fuel cost for different routes. It compares the costs and identifies the route with the lowest fuel expense.